

✓ Visualization:-

- Virtualizatⁿ is a technology that creates virtual version of computing resources such as servers, storage, and network.
- allows multiple virtual instances or environment to run on a physical system.

Characteristics:-

- Isolation
 - ensures separatiⁿ betⁿ VM.
- Resource pooling
 - Dynamically allocates resources based on demand
- Snapshot & cloning
 - Allows quick backups & replicatiⁿ of configurations.
- Dynamic resource allocatiⁿ:-
 - adjust resources dynamically for scalability.

Impact on Resource Utilizatiⁿ & Scalability

- Vir. positively impacts resource utilization by allows multiple VM virtual environment to share same physical hardware.
- This consolidatiⁿ leads to more efficient use of resources.
- Scalability is enhanced as virtual env (ve) can be easily scaled up or down based on demand.

How Vir enhances h/w utilization & reduce cost

→ 1) Consolidation

Multiple VM can run on a single physical system server, consolidating workloads & maximize use of h/w resources

2) Efficiency

enables better utilization of CPU, memory, & storage, minimizing wasted resources

3) Server Consolidation

Org can reduce no. of physical servers required, leads to lower h/w acquisition & maintenance costs

Illustrate char. of Google file system & how Google file system provides scalability?

⇒ GFS is a distributed file system developed by Google to provide scalable & reliable storage for large-scale distributed applicatⁿ.

It's designed to handle massive amts of data across thousands of commodity servers.

char:-

1) Scalability:-

- GFS designed to scale horizontally,
- allowing it to handle petabytes of data spread across thousands of servers.
- seamlessly add or remove storage nodes to accomodate changing storage requirements.

2) Fault-Tolerance

- It can continue to operate even in presence of hardware failures or network issues.
- achieves fault tolerance through data replicatⁿ & automatic recovery mechanisms.

3) chunk-based storage

- divides files into fixed-size chunks (typically 64 or 128 MB) & stores each chunk as a separate file on underlying storage nodes.

→ Simplified Namespace (S.N)

- uses SN where files are organized into a hierarchical directory structure similar to traditional file systems.
- Each file is identified by a globally unique file name.

(GFS)
Google File System

HDFS
Hadoop Distributed Sys

- | | |
|--|---|
| <ul style="list-style-type: none"> - Developed by Google - Primarily used by Google's infrastructure for data storage & processing. - uses master-slave architecture with a single master & multiple chunk servers. - doesn't use checksum - Not an open-source ecosystem | <ul style="list-style-type: none"> - Derived from Google's GFS paper but implemented by Apache Hadoop. - widely used in big data analytics and storage solutions. - Master-slave architecture with NameNode as master & multiple Data Nodes. - uses checksums - Open-source ecosystem. |
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GFS & HDFS Architecture & Differences

⇒ GFS Ar:-

1) Master slave A

follows master-slave arch. consists of a single master server & multiple chunk servers.

2) Master server (The Master):-

manages metadata about files & directories.

maintains inf about file locatⁿ, file chunk mappings & handles client req. for file operations.

3) Chunk servers:-

stores actual data in form of fixed-size data chunks. These servers are responsible for storing & serving data upon client req.

4) Client:-

interact with both master & chunk servers to read, write, & modify data.

HDFS Arch:-

- 1) Master-slave:- (msa)
follows msa, consists of single NameNode (master) & multiple DataNodes (slaves).
- 2) NameNode (The Master):-
manages file system namespace & metadata. It stores inf. about file locations, file permissions & directory structure.
- 3) DataNodes:-
stores actual data in the form of data blocks (default block size - 128 MB), responsible for serving & storing data upon client req..
- 4) Clients:-
interact with NameNode to retrieve metadata inf about files & directories & directly communicating with DataNodes to perform read & write operations on data blocks.

Diff:-

1) Terminology:-

while both sys have a master-slave arch, they use diff terminologies.

GFS uses master & chunk servers, whereas HDFS uses NameNode and DataNodes.

of Data Handling:-

GFS stores data in fixed-size chunks on chunk servers, whereas HDFS stores data in blocks on DataNodes.

HBase

DynamoDB

Origin -	Apache HBase, part of Hadoop ecosystem	- Amazon web services
-	Strong Eventual consistency	- offers both eventual reads & Strong consistency reads.
-	Horizontally Scalable	- Horizontally scalable.
-	Query lang is HBase Shell, Java API, REST API.	- AWS SDKs, AWS Management Console, & other client lib.
-	Supports atomic operat ⁿ at row level	- Supports atomic ops at record level.
-	requires manual setup, configurat ⁿ & maintenance	- fully managed by AWS
-	can be integrated with Hadoop ecosystem tools like Hadoop MapReduce & Apache Spark.	- integrates with AWS services such as AWS Lambda, S3, EMR.
-	Suitable for applicat ⁿ require ⁿ real-time, random read & write access.	- IOT, gaming, web & mobile apps, etc.

Cloud Storage Features:- SCAP

- refers to storage of data on remote servers accessed through internet.
- allows users to store & manage data without need for physical hardware & infrastructure.

Features.

1) Scalability:-

- offer scalable storage solutions
- Users can ~~sto~~ scale up or down their storage capacity as required without need for upfront investment.

2) Accessibility:-

- enables users to access their data from anywhere with an internet connection.

3) Durability:-

- are designed to ensure high durability of stored data.

4) Cost-effectiveness:-

- offers cost-effective solⁿ for storing large vol. of data compared to traditional on-premises storage solⁿ.

Various Types of Cloud Storage:- & compare

1) Object Storage:-

- Suitable for storing unstructured data such as images, vid, & documents.
- offers scalability, durability, & accessibility

Eg:- Amazon simple storage service (S3),
Google Cloud storage (GCS),
Azure Blob storage.

2) Block Storage:-

- provides block-level storage volumes that can be attached to virtual machines.
- suitable for databases, app data, & file sys.
- offers high performance.

3) File Storage:-

- offers shared file storage accessible via standard file protocols such as NFS & SMB.
- Suitable for file sharing, content mgmt,

Eg - Amazon Elastic File Sys (EFS),
Google file storage cloudstore Filestore,
Azure fdes

PAGE NO.
DATE

Object-based cloud flow object storage work & its drawbacks.

- It is a storage arch that manages data as objects rather than as traditional files systems.
- Each obj includes data itself metadata & a unique identifier.

Working:-

1) Object creation:-

- When user uploads its data to object-based cloud storage, the data broken down into obj.
- Each obj assigned a unique identifier & stored in storage system

2) Metadata Management:-

Along with data, obcs stores metadata associated with each object.

~~meta~~ Meta data provides inf about the object, such as its name, size, create date etc.

3) Storage Architecture:-

- Obcs sys typically use a distributed arch where data is stored across multiple nodes or servers in a cluster.
- Each node is responsible for storing & managing subset of objects.

~~Access~~

4) Access mechanism

- Obs's sys provides API's for accessing & managing objects.
- So that user can interact with storage system

Drawbacks

- 1) Limited Performance for small files.
- 2) Provide eventual consistency
- 3) may require certain level of expertise so it is complex.

Q. Various Cloud storage Provider.

- ⇒ 1) Amazon Web Services.
- AWS offers simple storage service (S3), which is scalable object storage service designed to store and retrieve any amt of data from anywhere on web.
 - provides highly durable and available storage infrastructure with data replication across multiple data centers.

2) Microsoft Azure:-

- Azure provides Azure Blob service storage, which is Microsoft's object storage service for storing and managing large amounts of unstructured data.
- Azure Blob storage offers high availability, scalability, & data redundancy.
- Users can interact using Azure Portal, Azure CLI, RESTful APIs.

3) Google cloud Platform (GCP)-

- Google cloud storage is Google's object storage service design to store & retrieve large amt of data.
- provides high durability, availability, performance etc.
- Users can access Google cloud storage through GCP console, client libraries.

4) IBM cloud:-

- IBM cloud offers IBM cloud Object storage, which is scalable, flexible object storage service for storing & managing unstructured data.
- User can access through IBM cloud console, RESTful APIs, SDKs, etc.

- 5) Alibaba cloud:- ^{scalable} Object Storage service (OSS) for storage
- multi-regional & zone redundant storage options.
 - Integration with Alibaba cloud like ECS, RDS & CDN.
 - Provides lifecycle management, & data backup.

Business Continuity :-

— refers to proactive planning and strategies adopted by org. to ensure that critical business func & operatⁿ can continue uninterrupted during & after disruption.

Cloud comput^g offers several adv for implement^g business continuity measures :-

- 1) Redundancy & Failover :-
cloud service providers often operate multiple data centers across diff. geographic regions.
Org.^s can leverage this distributed infrastructure to replicate their critical systems & data across multiple availability zones or regions.
- 2) Scalability & Elasticity :-
- allow org. to dynamically scale their resources up or down based on demand.
- 3) Data Replicatⁿ & Backup :-
cloud storage services offers built-in features for data replication, backup, & archival.

How BC is correlated with Disaster Recovery:-

⇒ Disaster recovery is procedure for restoring IT systems, application & data in the event of a disruptive incident or disaster.

DR & BC are correlated because:-

Shared Objectives-

Both in the cloud share obj of ensuring of continuous availability & access to critical data, application & services, even in face of disasters, or disruptions

Data replicatⁿ & Backup:-

offer capabilities for data replication, backup and disaster recovery across multiple regions or availability zones. Org can replicate data across data centers to ensure redundancy & availability.

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* DynamoDB

- ⇒ - NoSQL database service provided by AWS.
- high scalability, built-in security
 - wide range of use cases, from web & mobile apps to gaming & IoT applications.
 - Supports 2 main data models.

- Key-Value

- Data is stored & retrieved using primary key, which is unique identifier for each item.

- Document-

- allows storing nested data structures like JSON objects within each item.

* Security:- provides features like encryption, AWS - IAM (Identity & Access Management).

* Pricing - pay-as-go pricing model

* Integratⁿ:- integrates with AWS services such as AWS Lambda, Amazon S3, Amazon EMR etc.

* Use cases:- mobile & web applications, gaming, IoT, etc.

* Scalability.

✓ Role of Hypervision in Virtualization & distinguish Type 1 & Type 2

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Q Explain cloud based NAS devices contribute to adv of IaaS soln

→ Cloud based Network attached Storage (NAS) devices contribute to adv of IaaS.

• Scalability:-

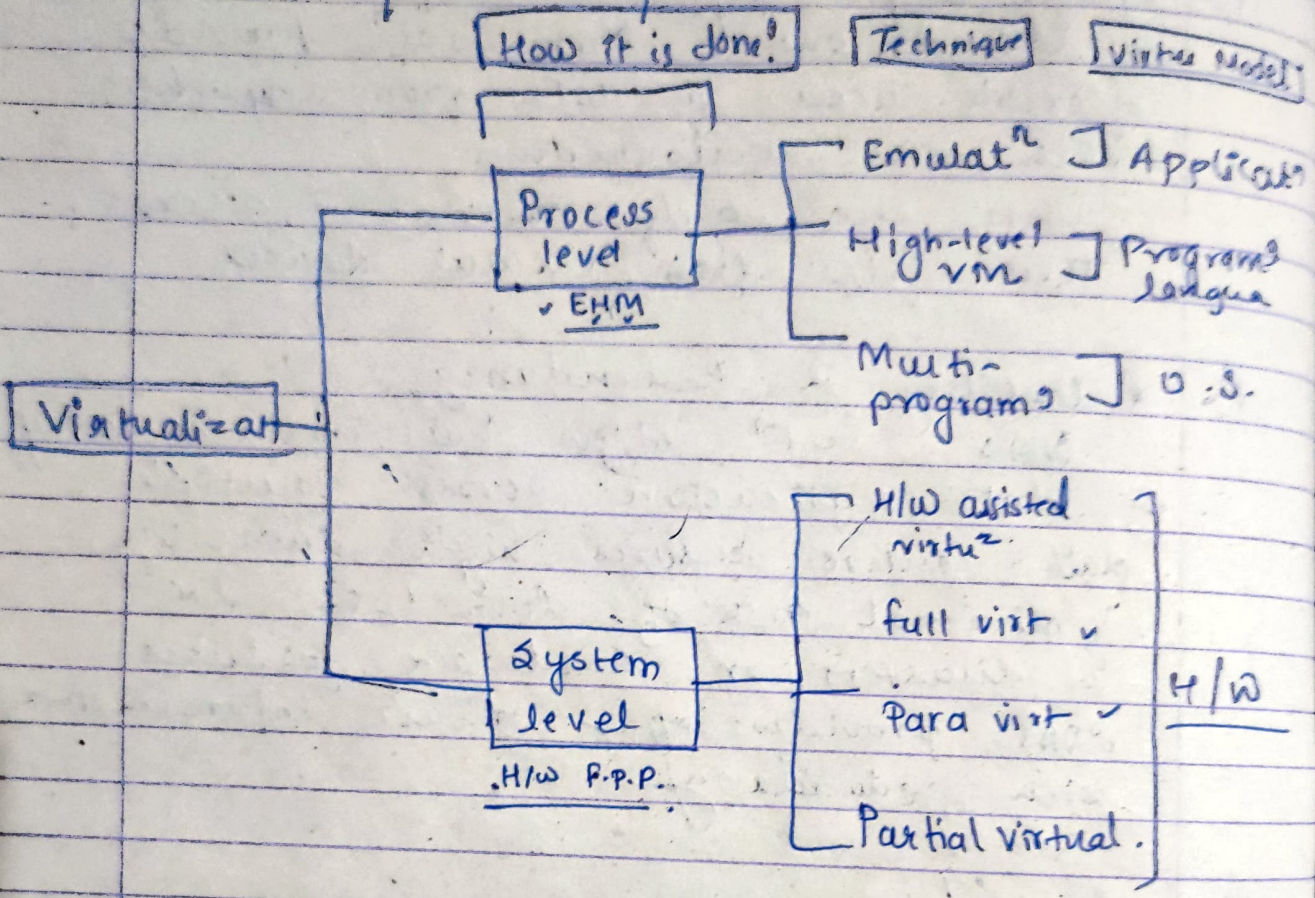
- offers scalability as users can easily scale up or down at their storage capacity without any need to for upfront investment.

• Cost-effectiveness

- With cloud based NAS, users pay for storage capacity that they need typically on a pay-as-you-go basis.
- This eliminates the need for large upfront investments in hardware allowing users to just pay for the resources they use rather than having physical infrastructure.

- 3) Flexibility & Accessibility:-
- Cloud based NAS solⁿ provide flexible access to data from anywhere with an ^{net} interconnection.
- Users can easily manage & access their files from various devices.
- 4) Reliability & Redundancy
NAS solⁿ offer built-in redundancy & data replication across multiple data centers. ensures high durability, & minimal risk of data loss due to disasters or hardware failures.
PaaS providers offer robust infrastructure with redundancy.
- 5) Security
cloud-based NAS solⁿ often include built-in security features such as encryption, access controls. This ensure confidentiality, integrity & availability of data storage.

Taxonomy of Virtualization:



Hardware Virtualization Technique:-

Full virtualization

- It refers to the ability of running a program, most likely on an O.S., directly on a top of virtual machines, without any modification.
- To achieve this, virtual machine manager are required to provide complete emulation of entire underlying H/W.

eg. VMware Esxi, Microsoft Hyper-V, KVM

• para-virtualization:-

- it involves modifying guest OS to make it aware of the virtualization layer.
- enables better performance & efficiency compared to full virtualizatⁿ.
- Eg. Oracle VMserver, Xen.

• Partial Virtualizatⁿ

certain

- It is where certain components of system are virtualized while others are not.
- Unlike para-vir, where guest OS is modified to interact with the hypervisor, in partial, guest OS typically runs unmodified.

• Execution Virtualizatⁿ

- includes technique that aim to emulate an execution environment that is separate from one hosting the virtualizatⁿ layer.
- can be implemented on top of hardware by OS, an applicatⁿ, libraries etc.